

Solving Two-Step Equations Using Properties of Equality

Teacher's Edition

Standard and Targets

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Boundary Forms are limited to $px \pm q = r$ and $p(x \pm q) = r$. The variable may be on either side, which is why Example 2 puts it there; worksheet question 11 is the related twist, with the constant printed before the variable term.

Learning targets I can relate properties of equality to solving two-step equations. I can solve two-step equations in one variable.

Essential question How can I apply the properties of equality to solve two-step equations?

Building on MA.6.AR.2.2 and MA.6.AR.2.3 — one-step equations with integer terms and solutions.

Working toward MA.8.AR.2.1 — multi-step linear equations with rational coefficients, including variables on both sides, and problems generating one solution, infinitely many solutions or no solution.

Where this sits in the course MA.7.AR.2.2 is required content of course 1205050, and these students have not taken grade 7 mathematics — they are meeting two-step equations **for the first time**, in the same year they sit the Grade 8 FAST. The code is not one the FAST reports.

MA.8.AR.2.1 is, and it is this benchmark with more steps and a letter on both sides; on the 2025 released items the Algebraic Reasoning benchmarks scored lowest of anything on the exam. There is no version of doing well there that goes around this lesson. Teach it as the first day of a skill, because that is what it is.

Materials Whiteboards and markers. Calculators. Worksheet 7A.01.03. Consider posting the Notes II table as a wall anchor chart — Math Nation suggests it and it earns its space for the rest of the year.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: spaced retrieval	2	4 min
Notes I–IV: what an equation says, the properties of equality, undoing layers, checking	3–6	13 min
Example 1: $5x - 9.6 = 21.4$, both steps and the check	7–9	5 min
Example 2: the variable on the right	10–12	5 min
Example 3: the one property that carries a condition	13–15	5 min
Whiteboards: 9 questions, each with its reveal	16–33	15 min

Segment	Slides	Time
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Notes II (slide 4) and Example 3 (slides 13–15). Notes II is the table students copy and use for the rest of the unit. Example 3 is the only place $c \neq 0$ gets a reason rather than an assertion.

Read this before you plan the day. Math Nation's own component times total 55 minutes — Warm-Up 5, Exploration 15, Bring It Together 5, Guided Instruction 15, Try It 10, Wrap-Up 5 — before any notes or worksheet. That does not fit the period. The plan above fits because the Exploration and Bring It Together are compressed into Notes II and Example 3: students still meet both tables and still meet division by zero, but as a copied anchor chart and a worked demonstration rather than a fifteen-minute discovery. The full Exploration needs a second day.

No partner or group work anywhere in this lesson. Math Nation's version has five, four of them in the Exploration and one opening the Try It. The Exploration tables are now Notes II and Example 3, taught from the front; the Try It is the whiteboard sequence plus the worksheet.

Vocabulary

equation — a statement that two expressions have the same value

solution — the value of the variable that makes the equation true

properties of equality — the moves that keep an equation true — symmetric, addition, subtraction, multiplication and division

coefficient — the number multiplied by the variable

Two notes on language. Math Nation lists no key terms for this lesson and five additional terms, all of them names of the properties. Those names are on the Notes II table rather than in the vocabulary bands, because a student needs the table to make sense of the name. And the division property is written here as it appears in Florida's own statement — **if $a = b$ and $c \neq 0$** — because the version without the condition is false, and this lesson spends a whole slide on why.

Slide-by-Slide Teaching Notes

Slide 1. Solving Two-Step Equations 1 min

- Post the learning target.
- Math Nation's own component times total 55 minutes before any notes or worksheet, which is why this deck compresses their Exploration and Bring It Together into Notes II and Example 3 — read the teacher's edition before you plan the day.
- The framing sentence is the one that matters: students arrive believing the equal sign means 'and the answer is', and everything today depends on replacing that.

Slide 2. Warm-Up 4 min

- TYPE: Spaced retrieval.
- INDEPENDENT.
- ANSWERS: 1) $45/99 = 5/11$. 2) 0.625 and 62.5%. 3) 10. 4) 24. Questions 3 and 4 look like filler and are not — every 'undo a subtraction' step today is question 3, and every negative coefficient is question 4. A student who is shaky on those will look shaky on equations, and it is worth knowing which problem you are actually looking at.
- OFF THE SLIDE, YOURS TO SAY: Questions 3 and 4 are the ones today's lesson leans on hardest.

Slide 3. Notes: I. An equal sign is a claim of balance 3 min

- THE IDEA THE WHOLE LESSON RESTS ON.
- Students arrive reading '=' as 'and the answer is', which is why $3 + 4 = 7 + 2 = 9$ looks acceptable to them.
- Write that chain on the board and ask whether 7 equals 9.
- The panel is the seesaw idea in words; draw the seesaw on the board once and refer back to it all week.
- OFF THE SLIDE, YOURS TO SAY: Whatever you do to one side, do to the other, or the balance breaks.

Slide 4. Notes: II. The properties of equality 4 min

- Students copy the table.
- Florida names these properties and the FAST uses their names, so the vocabulary is not optional.
- The symmetric row is the one students find pointless until whiteboard question 5, where an equation arrives with the variable on the right.
- OFF THE SLIDE, YOURS TO SAY: Only division carries a condition. You may never divide by zero.

Slide 5. Notes: III. Undo the outside layer first 3 min

- The order is the reverse of the order of operations, and saying that out loud helps: the equation was built by multiplying and then adding, so it is taken apart by subtracting and then dividing.
- Students who divide first are not wrong so much as slow — it works, and it produces fractions immediately.

- OFF THE SLIDE, YOURS TO SAY: Order matters. Peel the outside layer off before you reach the variable.

Slide 6. Notes: IV. Check by substituting into the ORIGINAL 3 min

- Worth its own slide because students who check at all tend to check against their own second line.
- Demonstrate it: make a deliberate sign error in step 1 at the board, then check against step 2 and watch it pass.
- The point lands in fifteen seconds and does not need repeating.
- OFF THE SLIDE, YOURS TO SAY: A mistake in step 1 passes a check done against step 2, every time.

Slide 7. Example 1 1 of the 5 min

- SPOKEN ONLY.
- Add 9.6 to both sides — the addition property of equality.
- Insist on the property name from the first example, or it will not survive to the worksheet.

Slide 8. Example 1 2 of the 5 min

- Write the property name beside each line as you go.
- The panel matters because the FAST accepts either route and students should not think one is the 'real' one.
- OFF THE SLIDE, YOURS TO SAY: Multiplying both sides by $\frac{1}{5}$ instead is the multiplication property. Also correct.

Slide 9. Example 1 2 of the 5 min

- Say the panel out loud.
- This is Notes IV done once in front of them, and every subsequent check this week should be held to the same standard.
- OFF THE SLIDE, YOURS TO SAY: Substituted into the equation we were given, not into the line $5x = 31$.

Slide 10. Example 2 1 of the 5 min

- Take a vote.
- It changes nothing at all — the symmetric property from Notes II says so — but students find that hard to believe and will often rewrite it first.
- Rewriting is allowed and it is worth saying that it is allowed rather than necessary.

Slide 11. Example 2 2 of the 5 min

- The last line is where the symmetric property earns its place in the table — you are allowed to turn $-32 = n$ around, and naming why is the difference between mathematics and habit.
- Watch for students who divide by 4 in the third line: n is already divided, so it has to be multiplied.

Slide 12. Example 2 2 of the 5 min

- The panel is the sign rule from the warm-up arriving in a division.

- A student who checks and gets 9.3 has read $-32 \div 4$ as $+8$; one who writes 6.7 has dropped the sign at the very end.
- Either way the check is what tells them.
- OFF THE SLIDE, YOURS TO SAY: The negative survives the division. $-32 \div 4$ is -8 , not 8.

Slide 13. Example 3 *2 of the 5 min*

- Whiteboards, the last column only.
- Rows 1 to 3 are quick and they are meant to be — the point of asking is that students commit to an answer for row 4 before they have thought about it.
- Take a show of hands on row 4 and write the count on the board; you will come back to it.

Slide 14. Example 3 *2 of the 5 min*

- Rows 1 and 2 are the reason the properties are worded the way Notes II words them: doing something to both sides is not the same as doing something to each side.
- Row 3 is the property working.
- Row 4 is not a false equation and students will call it one — it is the row with no number in it.
- OFF THE SLIDE, YOURS TO SAY: Rows 1 and 2 are why the properties say the SAME operation on both sides.

Slide 15. Example 3 *1 of the 5 min*

- Be precise here and do not let it become a slogan.
- Math Nation's answer key writes "undefined = 27" on this question, which cannot be right — you cannot put a non-number in an equation.
- Ask their own teacher tip's question out loud: how many times does 0 go into 13.5? Then answer it precisely, because the slogan version is false: NO number works, because 0 times anything is 0 and never 13.5.
- (Math Nation asks it about 56; the number is changed here so it matches the $5.5 + 8$ the rest of the example runs on.) (The 'any answer works' line belongs to $0 \div 0$, which is a different kind of undefined and is not this lesson.) Then point back at the $c \neq 0$ in Notes II and say that is where it came from.
- OFF THE SLIDE, YOURS TO SAY: Dividing by 0 does not give a false equation. It gives no number at all.

Slide 16. Whiteboards • Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards • Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline.
- Everything after this changes exactly one thing about it.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 18. Whiteboards · Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sign of the constant.
- The first move flips from subtract to add, and the answer more than doubles.
- A board saying 6 has run the first question again from memory.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 20. Whiteboards · Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sign of the coefficient.
- Same first move as question 1; the second move divides by a negative.
- Warm-up question 4 arriving on cue — a board saying 6 divided by 2 instead of -2 .
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 22. Whiteboards · Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the variable is divided, not multiplied.
- So the second move multiplies.
- A board saying 6 has divided out of habit — the same number as question 1, which is exactly why that board is worth finding.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sides are swapped.
- SAME ANSWER as question 4, and that is the whole point — the symmetric property from Notes II says the equation is unchanged.
- Anyone who got a different answer here changed something they were not allowed to change.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES IT.
- One feature changed: the two terms on the left have swapped places.
- Same answer a third time.
- Put questions 4, 5 and 6 up together and ask what actually changed between them — the arrangement, and nothing else.
- That is the properties of equality, seen rather than recited.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the coefficient is a fraction.
- Same two moves, same order — subtract 7 to get $\frac{2}{3}x = 12$, then undo the $\frac{2}{3}$.
- THE BOARD TO LOOK FOR SAYS 8, and it is two different errors landing on the same number: multiplying 12 by $\frac{2}{3}$ instead of dividing, and dividing 12 by 3 and then multiplying by 2 in the wrong order.
- Both are worth naming.
- The move that works is multiply by the reciprocal, $\frac{3}{2}$, and $12 \times \frac{3}{2} = 18$.
- Check it in the original before moving on — $\frac{2}{3}$ of 18 is 12.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.

- Do not reveal until every board is up.

Slide 31. Whiteboards • Question 8 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSSES IT.
- One feature changed: the coefficient is written as a decimal.
- SAME ANSWER as questions 4, 5 and 6 — the fourth time this round lands on 24 — because $0.5x$ and $x \div 2$ are the same instruction in different handwriting.
- THE BOARD TO LOOK FOR SAYS 6: that student multiplied by 0.5 instead of dividing, and 6 is question 1's answer, which is why it looks plausible.
- Put questions 1, 4 and 8 up together at the end.
- Three of them look different and two of them are the same equation.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 32. Whiteboards • Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards • Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: substituting gives $-2(6) + 7 = -5$, which is not 19, so she is wrong; the answer is $x = -6$, and $-2(-6) + 7 = 19$.
- MAYA'S ERROR IS QUESTION 3's, and it has a name: she divided by 2 instead of by -2 .
- The point of asking for the check rather than the answer is that the check is the part that works when nobody is watching — it caught an error here without knowing what the error was.
- A board that writes only " $x = -6$ " has the right answer to a question that asked for two things.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 34. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.03 — Answer Key

1. matching	Symmetric B • Addition D • Subtraction E • Multiplication A • Division C B is the symmetric property: $5 = 3 + 2$ becomes $3 + 2 = 5$, with nothing added, removed or scaled. This is the match students guess at.
2. describe $2 + 3n = 11$	multiplied • added • equal
3. isolate $3n$	Subtract 2 from both sides. Subtraction property of equality.
4. isolate n	Divide both sides by 3. Division property. Multiplying both sides by $1/3$ is equally correct and is the multiplication property.
5. solve $2 + 3n = 11$	$n = 3$ $3n = 9$, then $n = 3$. Check: $2 + 3(3) = 11$.
6. $2x - 5 = 11$	$x = 8$ Addition, then division. Check: $2(8) - 5 = 11$.
7. $n/12 - 3 = 7$	$n = 120$ Addition, then multiplication. Check: $120 \div 12 - 3 = 7$.
8. $6y + 5 = 29$	$y = 4$ Subtraction, then division. Check: $6(4) + 5 = 29$.
9. $9a + 7 = 88$	$a = 9$ Subtraction, then division. Check: $9(9) + 7 = 88$.
10. $y/4 - 8 = 4$	$y = 48$ Addition, then multiplication. Check: $48 \div 4 - 8 = 4$.
11. $-8 + b/3 = 10$	$b = 54$ Addition, then multiplication. The constant coming first changes nothing. Check: $-8 + 54 \div 3 = 10$.

Question 1 is the one Math Nation got wrong. Their version offers “ $5 = 3 + 2$ ” as the symmetric example — which is just the starting equation, unchanged. Students matched it by elimination and learned nothing. Here the symmetric example is $3 + 2 = 5$, so the two sides actually trade places. If a student can explain that match, they understand the property.

Accept more than one property name. Dividing both sides by 3 and multiplying both sides by $1/3$ are both correct and are different properties. Florida explicitly says not to mandate one solution path, and the sample answers above name the more common one only.

Question 11 is the one to read tonight. It puts the constant before the variable term and the variable over a denominator at the same time. A student who solves it has the method rather than a memorized shape, and that is what MA.8.AR.2.1 asks for at the end of Lesson 7, where the unknown arrives on both sides.

Additional Practice 7A.01.03 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. matching	Division B · Subtraction A · Multiplication D · Symmetric C · Addition E C is the symmetric property: $4 + 5 = 9$ becomes $9 = 4 + 5$.
2. describe $33t - 99 = 0$	multiplied · subtracted · equal
3. isolate $33t$	Add 99 to both sides. Addition property of equality.
4. isolate t	Divide both sides by 33. Division property of equality.
5. solve $33t - 99 = 0$	$t = 3$ $33t = 99$, then $t = 3$. A right side of 0 is not a special case. Check: $33(3) - 99 = 0$.
6. $-4 + x/2 = 16$	$x = 40$ Addition, then multiplication. Check: $-4 + 40 \div 2 = 16$.
7. $12n + 9 = 93$	$n = 7$ Subtraction, then division. Check: $12(7) + 9 = 93$.
8. $6x - 12 = 48$	$x = 10$ Addition, then division. Check: $6(10) - 12 = 48$.
9. $5y + 7 = 47$	$y = 8$ Subtraction, then division. Check: $5(8) + 7 = 47$.
10. $x/5 - 8 = 17$	$x = 125$ Addition, then multiplication. Check: $125 \div 5 - 8 = 17$.
11. $y/5.2 + 3.4 = 4.3$	$y = 4.68$ Subtraction, then multiplication. A decimal in the denominator changes nothing. Check: $4.68 \div 5.2 + 3.4 = 4.3$.

Misconceptions to Watch

- **Not realizing the variable $-x$ is multiplied by negative one.** Named in Math Nation's list for this lesson. (Math Nation TE)
- **Not seeing why multiplying by a reciprocal is efficient.** Also named in their list — and it is why Example 1 shows both routes. (Math Nation TE)
- **Trouble understanding why division by zero is undefined.** Named in their list, and the whole point of slides 13–15. (Math Nation TE)
- **Applying an inverse operation to only one side.** Florida names this for two-step equations. The seesaw drawing from Notes I is the antidote — one side goes down. (Florida B1G-M)
- **Misidentifying which number is the constant and which is the coefficient.** Also named by Florida. Worksheet question 2 is built to surface it. (Florida B1G-M)

- **Reading the equal sign as “and the answer is.”** Shows up as students solving only the left side. Notes I item B.
- **Thinking an answer written as $-32 = n$ is unfinished.** It is the symmetric property, and it is finished. Example 2 ends on exactly that line.
- **Believing the same operation on both sides always preserves equality.** It does not, if that operation is dividing by zero. This is why the property carries $c \neq 0$.

Changes from the Math Nation Lesson

- Every partner and group activity is replaced. There were five: two “work with your partner to complete the table” prompts and one “with your partner, compare” in the Exploration, a “discuss with your partner” in question 3a, and the Try It’s opening line.
- The symmetric property was corrected in three places. Math Nation’s teacher edition illustrates it with “If $2 = 2$, then $2 = 2$,” which is vacuous; the worksheet offers “ $5 = 3 + 2$ ” as the symmetric example, which is the unchanged starting equation; and the Additional Practice repeats the error with “ $4 + 5 = 9$.” In all three the two sides now actually trade places.
- The numeric examples in the properties table were rebuilt on $7 = 3 + 4$. Math Nation uses $a = b = 2$ for all five properties, so every example reads “if $2 = 2$ then $2 = 2$ ” with something done to both sides. A property about two expressions that look different needs two expressions that look different.
- The division property is stated with its condition. Math Nation’s own table has “if $a = b$ and $c \neq 0$,” but the summary statement students complete one page later says the same operation on both sides always preserves equality — which is false for division by zero. The notes here say the same nonzero number.
- The answer “undefined = 27” was removed. You cannot place a non-number inside an equation; dividing both sides by zero produces no equation at all. The row is kept, because the discussion it starts is the reason the division property carries a condition.
- Worksheet question 1 was rebuilt as a lettered matching item. As printed it was two side-by-side tables with no letters, no numbers and no blanks, so there was no way for a student to record an answer or for you to mark one. The new format matches the FAST matching-item style.
- A solve step was added. Questions 2 through 4 set up $2 + 3n = 11$, describe it, and then abandon it — the next question jumps to a different equation. Question 5 now finishes it.
- Three consecutive answers of 7 were broken up. On the Additional Practice, $12n + 9 = 93$, $6x - 12 = 30$ and $5y + 7 = 42$ all gave 7. Two of the three were retuned to give 10 and 8.
- The Wrap-Up was replaced. Math Nation’s Cool Down asks two explanation questions and keys both as “Answers will vary,” so it produces no evidence about who can solve an equation. The Ticket Out the Door has students solve one, name the properties, and explain the $c \neq 0$ condition.
- Notes I keeps the balance idea and the Warm-Up drops the optional video. Balance is genuinely the mathematics here, so it stays — as a drawing you make at the board, since nothing on these slides is a picture; the video does not fit a lesson already over the period.
- Math Nation’s instruction to avoid the word “cancel” was kept and promoted into the slide notes. It matches Florida’s guidance and it is the difference between naming a property and waving at one.

- Answer keys were built for both sheets, from the same source file as the student versions, so the two cannot drift apart.
- Nothing in the mathematics was changed apart from the symmetric-property corrections and the two retuned values noted above.